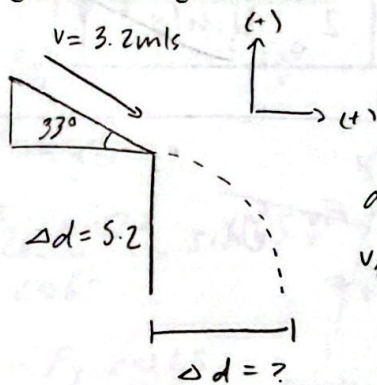


	4+	Level 4 8-9 marks	Level 3 7 marks	Level 2 6 marks	Level 1 5 mark	NOT YET
Thinking/ Inquiry Questions answered using the formulas and units required to demonstrate understanding of the concepts.	10 marks All questions answered using the correct formulas and units as required. A thorough and correct understanding of All concepts/ questions demonstrated.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of Most concepts/ questions demonstrated.	All questions answered using most of the correct formulas. A good understanding of most concepts/ questions demonstrated.	Most questions answered or attempted. Questions answered using some of the correct formulas. Some understanding of concepts/ questions demonstrated.	Some questions answered or attempted. Questions answered using limited formulas. Limited understanding of concepts/ questions demonstrated.	Questions not answered or show little understanding of concepts.

For full marks, Clearly show ALL of your work for every question. Include FBD's, FOR, Fnet statements (x and y), units, and directions where appropriate. Round your final answer to 3 sig figs.

Thinking

1. A child throws a ball onto the roof of a house, then catches it with a baseball glove 1.0m above the ground as seen in the figure to the right. The ball leaves the roof with a velocity of 3.2 m/s. What is the horizontal distance from the glove to the edge of the roof.



$$\begin{aligned}
 v &= 3.2 \text{ m/s} \\
 a &= -9.8 \text{ m/s}^2 \uparrow \\
 v_{iy} &= -3.2 \text{ m/s} \uparrow \cdot \sin 33 \\
 &= -1.7428 \text{ m/s} \uparrow \\
 \Delta d &= -5.2 \text{ m} \uparrow
 \end{aligned}$$

$$t = ?$$

$$\begin{aligned}
 \Delta d &= v_{iy}t + \frac{1}{2}at^2 \\
 \frac{1}{2}(-9.8)t^2 - 1.7428t + 5.2
 \end{aligned}$$

quad. formula

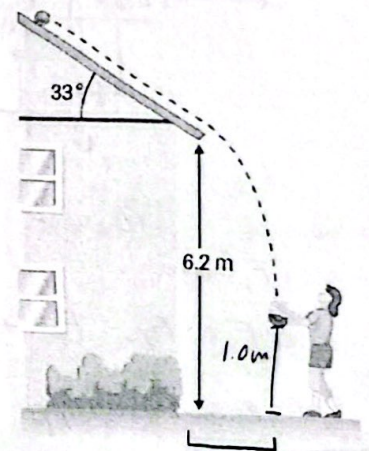
$$t = 0.86756 \text{ s} \quad \text{or} \quad 1.2252$$

$$t = 0.86756 \text{ s}$$

$$\begin{aligned}
 v_x &= 3.2 \text{ m/s} \rightarrow \cdot \cos 33 \\
 &= 2.6837 \text{ m/s} \rightarrow
 \end{aligned}$$

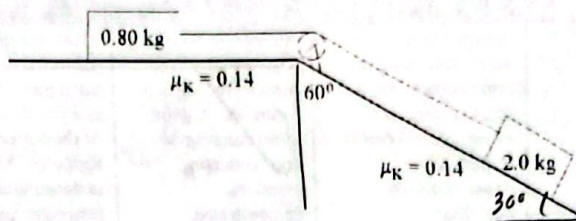
$$d = vt$$

$$\begin{aligned}
 d &= 2.6837 \text{ m/s} \rightarrow \cdot 0.86756 \text{ s} \\
 &= 2.33 \text{ m} \rightarrow
 \end{aligned}$$

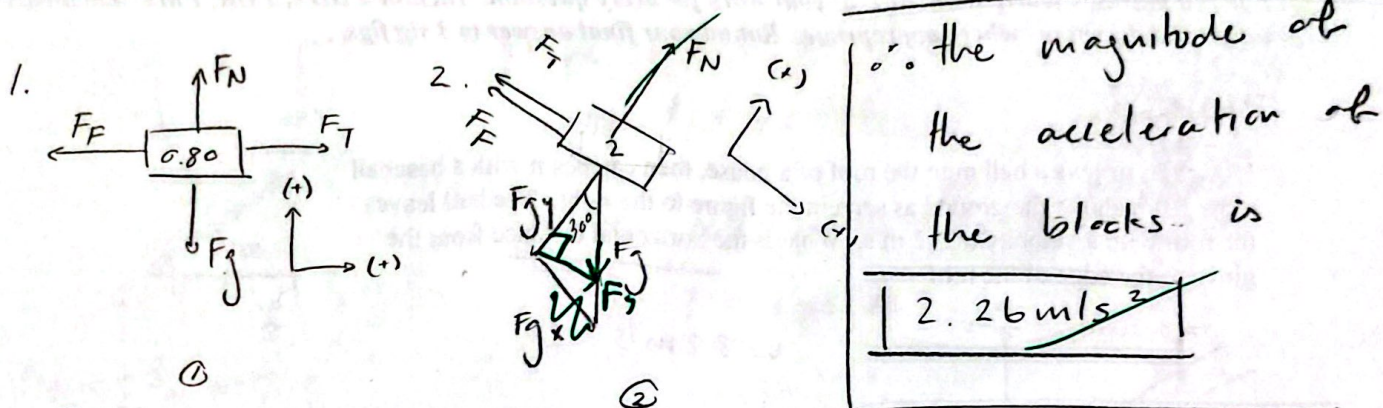


∴ the horizontal distance from the glove to the edge of the roof is 2.33m [left]

2. Two blocks are connected to each other through a pulley as shown in the diagram below. The block on the table has a mass of 0.80 kg and the block on the 30 degree ramp has a mass of 2.0 kg. The coefficient of friction is 0.14 for each block.



- a) Draw an FBD and FOR for each block. Determine the magnitude of acceleration of the blocks. CLEARLY Show ALL of your.



$$F = ma$$

$$\sum F = F_T - F_F + F_N - F_g$$

$$m_1 a = F_T - F_{1F}$$

$$F_T = m_1 a + F_{1F}$$

$$F = ma$$

$$F = F_{Tx} - F_T - F_F + F_N - F_{gy}$$

$$m_2 a = F_{2gx} - F_T - F_{2F}$$

$$F_T = F_{2gx} - F_{2F} - m_2 a$$

$$m_1 a + F_{1F} = F_{2gx} - F_{2F} - m_2 a$$

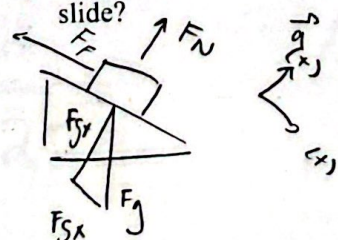
$$a = \frac{F_{2gx} - F_{2F} - F_{1F}}{m_2 + m_1}$$

$$(0.8 \text{ kg})(9.8 \text{ m/s}^2)(1)(0.14)$$

$$a = \frac{(9.8 \text{ m/s}^2)(1)(2 \text{ kg})(\sin 30) - (9.8 \text{ m/s}^2)(1)(2 \text{ kg})(\cos 30)(0.14) - (0.8 \text{ kg})(9.8 \text{ m/s}^2)(1)(0.14)}{2.8 \text{ kg}}$$

$$= \frac{6.3260}{2.8 \text{ kg}} = 2.2593 \text{ m/s}^2$$

- b) If the string broke, for what minimum value of the coefficient of static friction would the 2.0 kg block not begin to slide?



$$\sum F = F_{gx} - F_F + F_N - F_{gy}$$

$$F_F = F_{gx}$$

$$(F_N)(\mu_s) = mg \sin 30$$

$$\mu_s = \frac{mg \sin 30}{mg \cos 30}$$

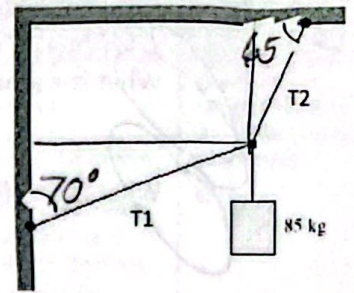
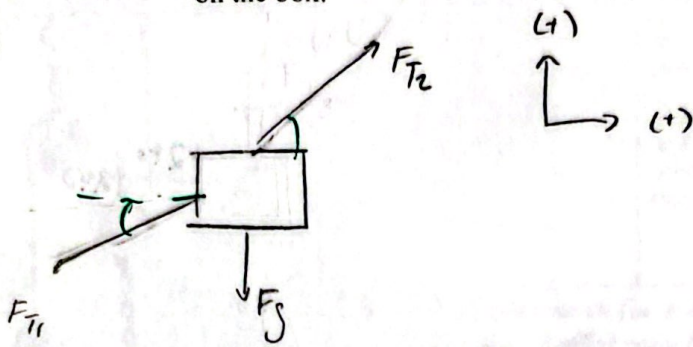
$$\mu_s = \tan 30$$

$$= 0.57735$$

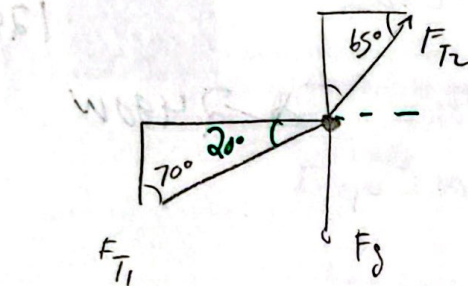
$$\therefore \text{the min value at } \mu_s = 0.577$$

3. An 85 kg object is suspended from a ceiling and attached to a wall as shown in the diagram. Calculate the tension in each rope.

- a. Considering the forces acting on the box, draw a FBD for the forces acting on the box.



- b. Calculate the tension in each rope. 4



$$\sum F_y = F_{T2y} - F_g - F_{T1y}$$

$$F_{T2y} = F_g - F_{T1y}$$

$$\textcircled{1} F_{T2} \sin 65 = mg - F_{T1} \cos 65$$

$$\sim \frac{2}{5} E$$

$$\times$$

$$\sum F_x = F_{T2x} - F_{T1x}$$

$$F_{T2x} = F_{T1x}$$

$$F_{T2} \cos 65 = F_{T1} \sin 70$$

$$\textcircled{2} F_{T2} = \frac{F_{T1} \sin 70}{\cos 65}$$

$$\textcircled{2} \rightarrow \textcircled{1}$$

$$\frac{F_{T1} \sin 70}{\cos 65} \cdot \sin 65 =$$

$$mg - F_{T1} \cos 65$$

$$F_{T1} \cdot \sin 70 \cdot \tan 65 + F_{T1} \cos 65$$

$$= mg$$

$$F_{T1} = \frac{mg}{\sin 70 \cdot \tan 65 + \cos 65}$$

$$F_{T1} = (85)(9.8)$$

$$\sin 70 \cdot \tan 65 + \cos 65$$

$$F_{T1} = 341.70 \text{ N} \quad 498 \text{ W}$$

$$F_{T2} = \frac{F_{T1} \sin 70}{\cos 65}$$

$$F_{T2} = \frac{(341.70 \text{ N})(\sin 70)}{(\cos 65)} = 759.79 \text{ N} \quad 20^\circ \downarrow$$

$\therefore$ the tension in $T_1 = 342 \text{ N} [E 25^\circ N]$
the tension in $T_2 = 760 \text{ N} [W 20^\circ S]$

4. A man attempts to raise a heavy crate by pulling a rope while exerting a pulling force of 1240 N Down.

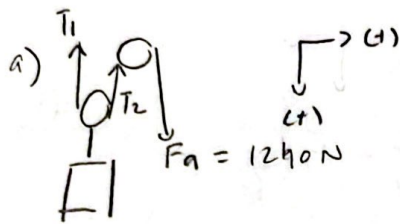
a. What is the Tension in Rope 1 (T_1)

1240 N

b. What is the Tension in Rope 2 (T_2)

1240 N

c. How heavy is the crate in kg?



a) the tension in rope 1 = 620 N [up]

b) the tension in rope 2 = 620 N [up]

$$\sum F_{NET} = F_a - T_2 - T_1$$

$$F_a = T_2 + T_1$$

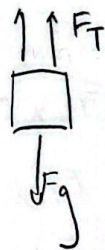
$$F_a = 2T$$

$$T = \frac{F_a}{2}$$

$$T = \frac{1240 \text{ N}}{2}$$

$$= 620 \text{ N}$$

$$c) F = ma = F_a - T_2 - T_1 + mg$$



$$mg = F_{T1} + F_{T2}$$

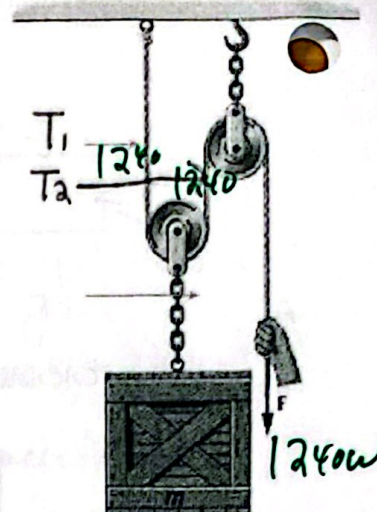
$$mg = 2F_T$$

$$m = \frac{2F_T}{g}$$

$$m = \frac{2(620 \text{ N} \times 2)}{9.8 \text{ m/s}^2}$$

$$9.8 \text{ m/s}^2 (\downarrow)$$

$$= 126.53 \text{ kg}$$



2480 N

$$F_{T1} = 620 \text{ N [up]}$$

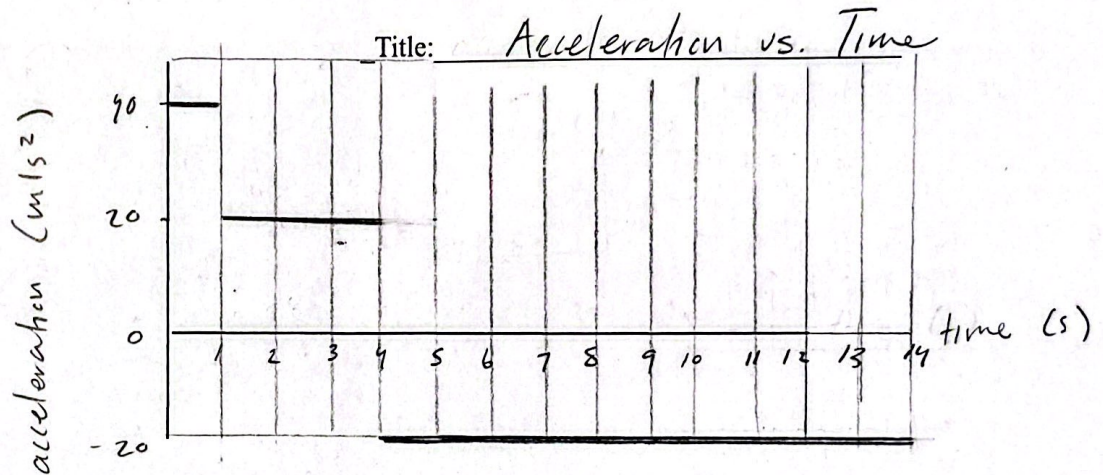
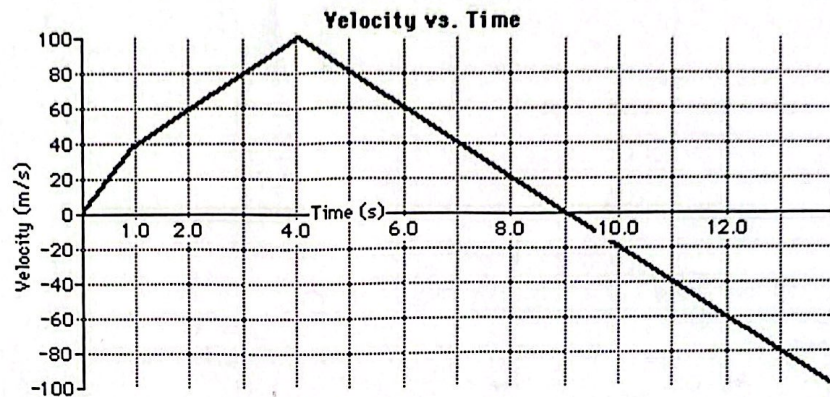
$$F_{T2} = 620 \text{ N [up]}$$

$$m = 127 \text{ kg}$$

$$253 \text{ kg}$$

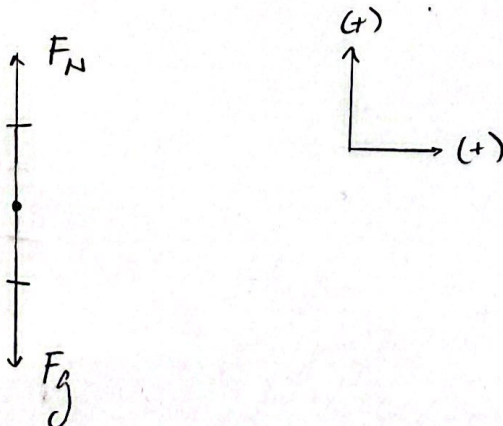
Communication (10 mark):

1. Plot an acceleration time graph for the object; make sure your lines or curves are clearly shown. Include a proper TITLE and Labels of the axis (3 marks)

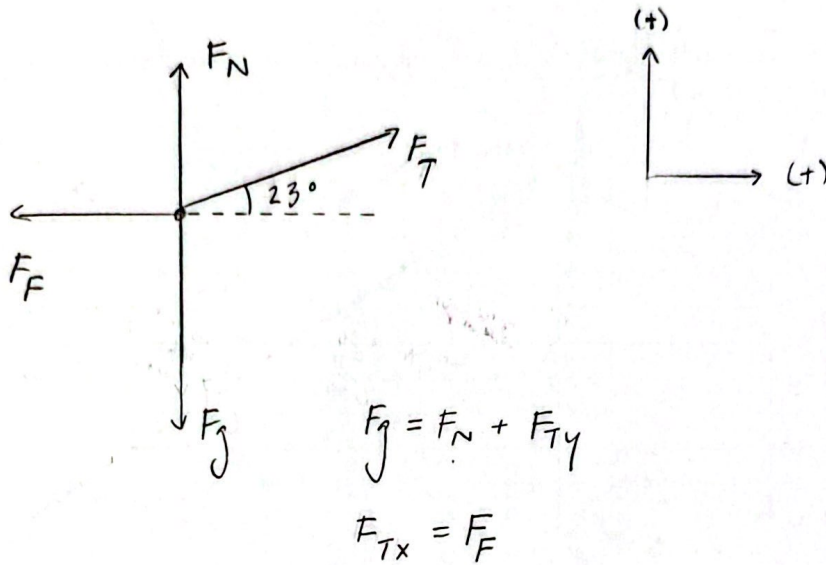


2. CLEARLY draw an FBD and FOR for the following scenarios. Be sure to label all appropriate forces. Relative lengths must be obvious!

- a) A hockey puck moving right at a constant velocity on ice with NO friction, After being hit with a hockey stick. (2 mark)



- b) A box full of textbooks being dragged across the floor at constant velocity, with friction, by a person pulling on a rope at an angle of 23° above the horizontal. (3 marks)



- d) Draw a FBD for the forces "tension from strings" acting on the mass. (2 marks)

